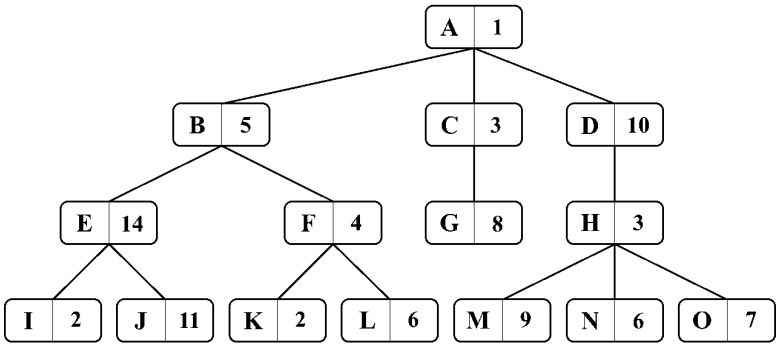
Problem D
Party Happiness Index

Time limit: **1 second**
Memory limit: **1024 megabytes**

Problem Description

Professor Stewart is consulting for the president of a corporation that is planning a company party. The company has a hierarchical structure; that is, the supervisor relation forms a tree rooted at the president. The personnel office has ranked each employee with a conviviality ranking, which is a real number. In order to make the party fun for all attendees, the president does not want both an employee and his or her immediate supervisor to attend.



Professor Stewart is given a tree that describes the structure of the corporation, shown as the above. Each node of the tree holds, in addition to the pointers, the name of an employee and that employee's conviviality ranking. We can make up a guest list that maximizes the sum of the conviviality ranking of the guests via solving the problem of maximum weighted independent set.

Please write a program to make up a guest list that maximizes the sum of the conviviality ratings of the guest.

Input Format

The first line of input contains an integer N which means the number of employees. Each of the subsequent N lines are composed of the employee's name S and happiness index R. After that, $N-1$ lines that describe the hierarchical structure. Each line of the tree specification has the form:

$L\ K$

It means that the employee L is an immediate supervisor of the employee K.There is a blank between L and K. Input is ended with the line '0'.

Output Format

Your program is to write to standard output. Print exactly one line for each test case. The line is an integer which represents the maximum sum of the conviviality ratings of the guest.

Technical Specification

- $1 \leq N \leq 6,000$
- $1 \leq R \leq 10,000$
- $|S| \leq 20$

Sample Input 1

```
15
A 1
B 5
C 3
D 10
E 14
F 4
G 8
H 3
I 2
J 11
K 2
L 6
M 9
N 6
O 7
A B
A C
A D
B E
B F
C G
D H
E I
E J
F K
F L
H M
H N
H O
```

Sample Output 1

```
66
```